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1.2.6 Equivalence of vector norms

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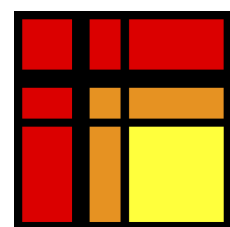
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1/1 point (graded)



Read Unit 1.2.6 of the notes. [[LINK](#)]

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? What is meant by the notation $||x||$?

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In Theorem 1.2.6.1, you introduce the term $\| \cdot \|$. Is this a new type or norm distinct from the $\| \cdot \|$ norms, or is it the norm of the absolute value...

? Could you elaborate more on what you meant by "if a vector is "small" in one norm, it is "small" in all other norms"?
Remark 1.2.6.3 says the following: > *The bottom line is that, modulo a constant factor, if a vector is "small" in one norm, it is "small" in all other...

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Homework 1.2.6.1

1.0/1.0 point (graded)
Fill out the following table:

Example: enter $4\sqrt{2}$ as 4 * sqrt(2)

x	$\ x\ _1$	$\ x\ _\infty$	$\ x\ _2$
$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$	1 Answer: 1	1 Answer: 1	1 Answer: 1
$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$	3 Answer: 3	1 Answer: 1	sqrt(3) Answer: sqrt(3)
$\begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$	4 Answer: 4	2 Answer: 2	sqrt(6) Answer: sqrt(6)

Solution:

For a complete solution see the [notes](#).

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Homework 1.2.6.2

1/1 point (graded)
Complete the proof of Theorem 1.2.6.1 in the notes.

done



Solution:

For a complete solution see the [notes](#).

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Homework 1.2.6.3

1/1 point (graded)
Let $x \in \mathbb{C}^m$. The following table organizes the various bounds:

	$\ x\ _1 \leq C_{1,2}\ x\ _2$	$\ x\ _1 \leq C_{1,\infty}\ x\ _\infty$
$\ x\ _2 \leq C_{2,1}\ x\ _1$		$\ x\ _2 \leq C_{2,\infty}\ x\ _\infty$
$\ x\ _\infty \leq C_{\infty,1}\ x\ _1$	$\ x\ _\infty \leq C_{\infty,2}\ x\ _2$	

For each, determine the constant $C_{x,y}$ and prove the inequality, including that it is a tight inequality.

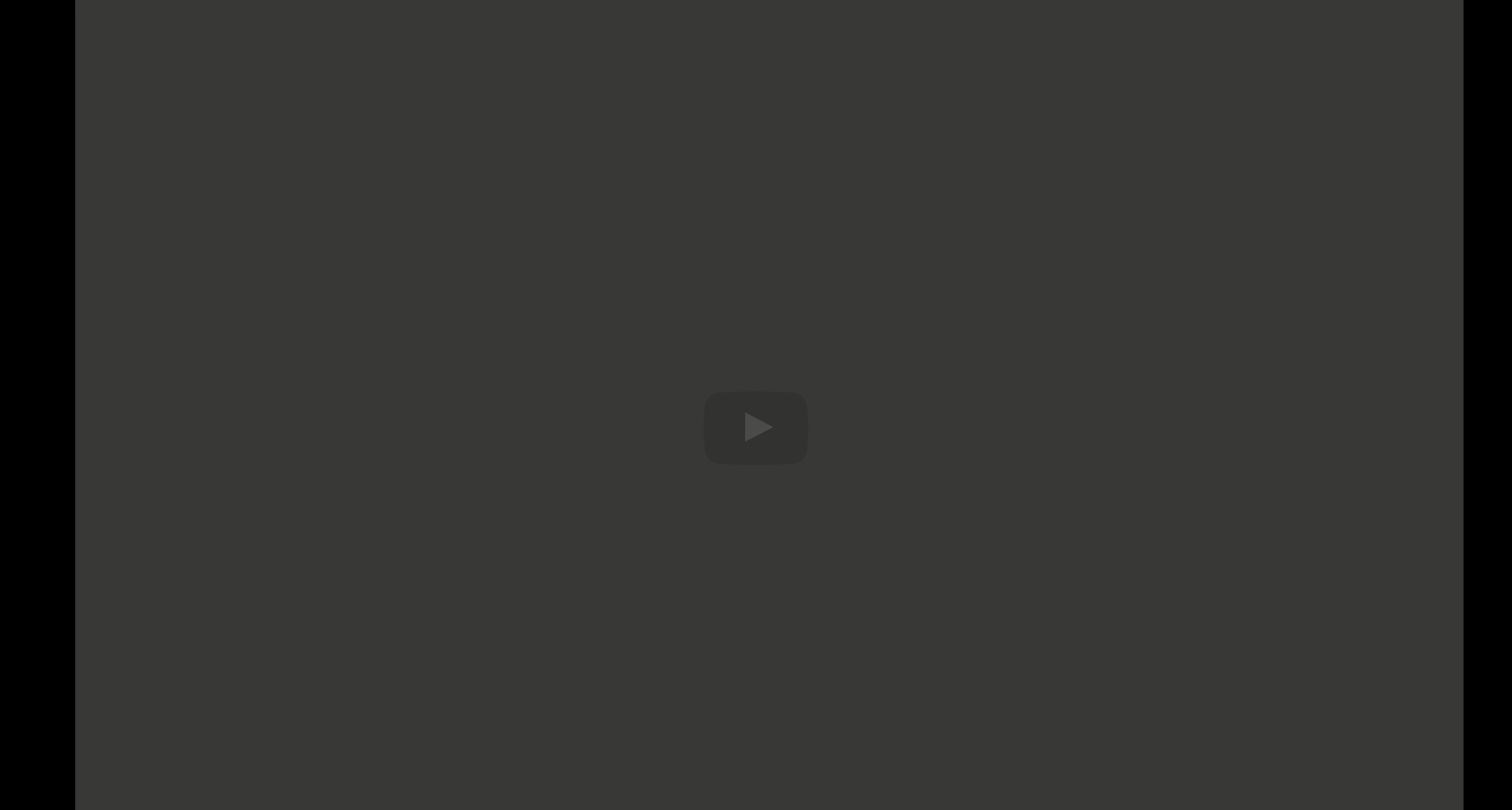
$C_{1,2} =$ ☐ 1 ☒ $\sqrt{m}$ ☐ m ✓	$C_{1,\infty} =$ ☐ 1 ☐ $\sqrt{m}$ ☒ m ✓
$C_{2,1} =$ ☒ 1 ☐ $\sqrt{m}$ ☐ m ✓	$C_{2,\infty} =$ ☐ 1 ☒ $\sqrt{m}$ ☐ m ✓
$C_{\infty,1} =$ ☒ 1 ☐ $\sqrt{m}$ ☐ m ✓	$C_{\infty,2} =$ ☒ 1 ☐ $\sqrt{m}$ ☐ m ✓

Full solution: See notes. [[LINK to unit](#)]

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▶ 2.0x

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